

SIMATS ENGINEERING

ASSESSMENT TOOL 1

COURSE NAME: Artificial Intelligence

COURSE CODE: CSA17

COURSE OUTCOME COVERED

CO1: Apply AI basics such as intelligent agents and uninformed search methods to solve problems effectively. (BL3)

Assessment Tool Weightage: 50%

QUESTIONS: 5

Duration: 1 Hour

Total Marks:50

Annexure A

Analytical Problem Solving

Code of Conduct:

*I **HEMASRINITHA P. (192572179)** (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.*

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

1. Solve the Water Jug problem: you are given 2 jugs, a 4-gallon one and 3-gallon one. Neither has any measuring maker on it. There is a pump that can be used to fill the jugs with water. How can you get exactly 2 gallons of water into 4-gallon jug? Explicit assumptions: A jug can be filled from the pump, water can be poured out of a jug onto the ground, water can be poured from one jug to another and that there are no other measuring devices available.

Aim

To solve the Water Jug Problem using the 4-gallon and 3-gallon jugs and obtain exactly 2 gallons in the 4-gallon jug.

Algorithm

1. Fill the 3-gallon jug completely.
2. Pour the 3 gallons into the 4-gallon jug.
3. Fill the 3-gallon jug again.
4. Pour water from the 3-gallon jug into the 4-gallon jug until it becomes full.
 - Only 1 gallon is needed to fill the 4-gallon jug.
 - So 2 gallons remain in the 3-gallon jug.
5. Empty the 4-gallon jug.
6. Pour the 2 gallons from the 3-gallon jug into the 4-gallon jug.

State Space Representation

Step	Action	4-Gallon Jug	3-Gallon Jug
Start	Initial state	0	0
1	Fill 3-gallon jug	0	3
2	Pour 3 → 4	3	0
3	Fill 3-gallon jug	3	3
4	Pour 3 → 4 until full	4	2
5	Empty 4-gallon jug	0	2
6	Pour 2 → 4	2	0

Goal State

- **4-gallon jug = 2 gallons**
- **3-gallon jug = 0 gallons**

Result: Exactly **2 gallons** of water are obtained in the **4-gallon jug**.

2. Find out how Mars Rover, analyse and explore the samples on Mars by answering the following questions.

i. What are the percept's for this agent?

ii. Characterize the operating environment.

iii. What are the actions the agent can take?

iv. How can one evaluate the performance of the agent?

v. What sort of agent architecture do you think is most suitable for this agent?

Why?

i. Percepts

The Mars Rover collects information from its sensors, such as:

- Images from cameras
- Rock and soil composition
- Temperature and weather conditions
- Terrain and obstacles
- Distance to objects
- Position and battery level

ii. Operating Environment

The Mars Rover operates in a partially observable, sequential, dynamic, continuous, and single-agent environment. It cannot observe the entire environment at once, and each action affects future actions. The terrain and weather on Mars may change over time.

iii. Actions

The Mars Rover can:

- Move in different directions
- Capture images
- Collect and analyze rock and soil samples
- Avoid obstacles
- Send scientific data to Earth
- Manage its power and stop when required

iv. Performance Evaluation

The rover's performance is evaluated based on:

- Accurate sample collection and analysis

- Successful exploration of target areas
- Safe navigation without collisions
- Efficient battery usage
- Reliable communication with Earth
- Number of useful scientific discoveries

v. Suitable Agent Architecture

A Goal-Based Agent is the most suitable architecture because the Mars Rover has specific goals such as exploring Mars, collecting samples, and analyzing them. It plans its actions to achieve these goals while avoiding obstacles and using resources efficiently.

3. Explore the search space using search strategies and formulate the problem components for the 8-Queens problem using the following information: Place 8 queens on a chessboard such that none of the queen's attack any of the others. A configuration of 8 queens on the board as shown in the figure below, but this does not represent a solution as the queen in the first column is on the same diagonal as the queen in the last column.

Problem Statement

Place 8 queens on an 8×8 chessboard such that no two queens attack each other. A queen can attack another queen if they are in the same row, same column, or same diagonal.

Problem Components

1. Initial State

- An empty 8×8 chessboard with no queens placed.

2. State Space

- All possible arrangements of 8 queens on the chessboard.
- Each state represents a different placement of queens.

3. Actions (Operators)

- Place one queen in a safe position on the next column.
- Move to the next column and repeat until all 8 queens are placed.
- If no safe position is found, backtrack and try another position.

4. Goal State

- Exactly 8 queens are placed.

- No two queens are in the same row.
- No two queens are in the same column.
- No two queens are on the same diagonal.

5. Path Cost

- Each queen placement has a cost of 1.
- The total cost is the number of moves made to reach the solution.

Search Strategy: Backtracking Search is used. It places queens column by column and backtracks whenever a conflict occurs.

Performance Measure

- All 8 queens are placed successfully.
- No queens attack each other.
- The solution is found with minimum backtracking and search time.

4. A customer wants to travel from the one location to another location using OLA Cab booking mobile application. The customer can select the any of the cab type as mini, micro, sedan, shared, prime and so on based on his comfort to reach the destination. Identify what type of problem-solving agent can be used and write the pseudocode for the above problem.

Type of Agent

A Goal-Based Agent is used because the customer's goal is to reach the destination by selecting the most suitable cab (Mini, Micro, Sedan, Shared, Prime, etc.) based on comfort, cost, and availability.

Pseudocode

START

Input pickup location

Input destination

Display available cab types

User selects a cab

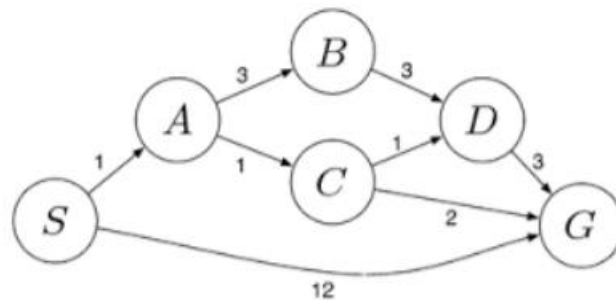
Check cab availability

```
IF cab is available THEN
    Book the cab
    Display driver details and fare
ELSE
    Display "Cab not available"
ENDIF
END
```

Conclusion:

A **Goal-Based Agent** is suitable because it selects the best action (booking the appropriate cab) to achieve the user's goal of reaching the destination.

5. A logistics company operates a delivery network where packages must be transported between warehouses at minimum cost. Each route between warehouses has an associated travel cost (fuel + time). The company wants to determine the least-cost path from the starting warehouse S to the destination warehouse G using the Uniform Cost Search (UCS) algorithm. The delivery network is represented as a weighted graph:



Algorithm

1. Start at node S with cost 0.
2. Add all neighboring nodes to the priority queue based on cumulative cost.
3. Remove the node with the lowest cost.
4. Repeat until the goal node G is reached.
5. The first time G is removed from the queue, the least-cost path is found.

UCS Traversal:-

Step	Expanded Node	Path Cost
1	S	0
2	A	1
3	C	2
4	B	4
5	D	3
6	G	4

Cost Calculation

Possible paths:

- $S \rightarrow G = 12$
- $S \rightarrow A \rightarrow B \rightarrow D \rightarrow G = 1 + 3 + 3 + 3 = 10$
- $S \rightarrow A \rightarrow C \rightarrow G = 1 + 1 + 2 = 4$
- $S \rightarrow A \rightarrow C \rightarrow D \rightarrow G = 1 + 1 + 1 + 3 = 6$

Least-Cost Path

$S \rightarrow A \rightarrow C \rightarrow G$

Total Cost = 4

Conclusion

Using the Uniform Cost Search (UCS) algorithm, the optimal delivery route is:

$S \rightarrow A \rightarrow C \rightarrow G$ with the minimum travel cost of 4.